File permissions in Linux

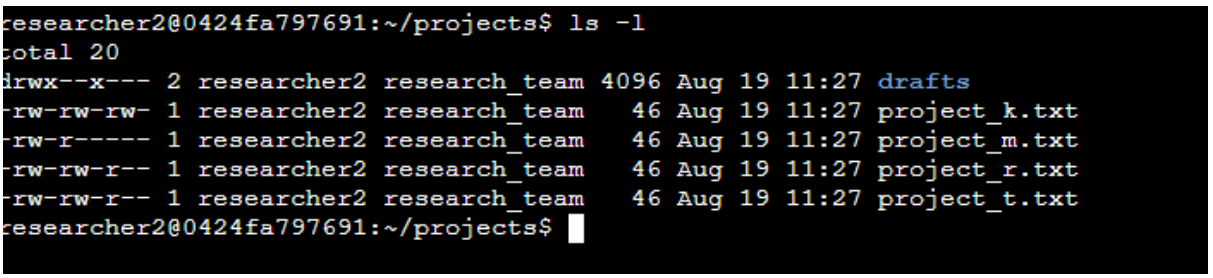
Project description

The research team at my organization needs to update the file permissions for certain files and directories within the projects directory. The permissions do not currently reflect the level of authorization that should be given. Checking and updating these permissions will help keep their system secure.

I used Linux commands to configure authorization. In that way I learned that setting appropriate access permissions is critical to protecting sensitive information and maintaining the overall security of a system. I also learned how to do that. Like chmod for changing permissions and ls -la for verify all permissions even in the occult files.

Check file and directory details

To list the contents and permissions of the `projects` directory. We should use **ls -l** or **ls -la** for showing occult files and all permissions.
The output should be like.



Describe the permissions string

If we get the first example from the previous image and check the drafts permissions we can have.

```
drwx--x---.
That means the user has all the permissions and the group has
executable permissions.
What indicates the hyphen -> That means the permission is not granted
or for the first spot means that it is a normal file.
The d -> means is a directory
2nd-4th-> user permissions
5th-7th-> group permissions
8th-10th-> other permissions
```

```
r-> read permission
w-> write permission
x-> execute permission
```

Change file permissions

In the `chmod` command, `u` sets the permissions for the user who owns the file, `g` sets the permissions for the group that owns the file, and `o` sets the permissions for others.

The `project_k.txt` file has write permissions for other users.

So we need to change that with the chmod command. using option o-w

```
researcher2@0424fa797691:~/projects$ chmod o-w project_k.txt
researcher2@0424fa797691:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Aug 19 11:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Aug 19 11:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Aug 19 11:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug 19 11:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Aug 19 11:27 project_t.txt
researcher2@0424fa797691:~/projects$
```

Change file permissions on a hidden file

Question -> The research team has archived **.project_x.txt**, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file. Use a Linux command to assign **.project_x.txt** the appropriate authorization.

We can use ls -la vor verify the permissions

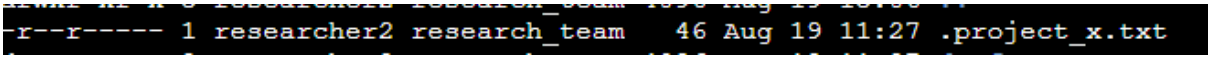
```
researcher2@0424fa797691:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 19 11:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 19 13:06 ..
-rw--w---- 1 researcher2 research_team   46 Aug 19 11:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 19 11:27 drafts
```

After showing up the permissions we need to change the write permissions from users and group, and add read permissions for group.

So we use chmod command like this - > **chmod u-w,g-w,g+r .project_x.txt**

```
researcher2@0424fa797691:~/projects$ chmod u-w,g-w,g+r .project_x.txt
```

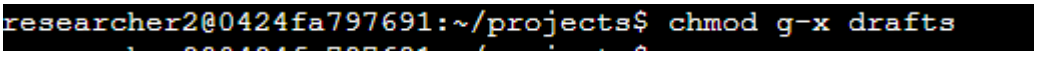
The file should change the permissions and stay like this.



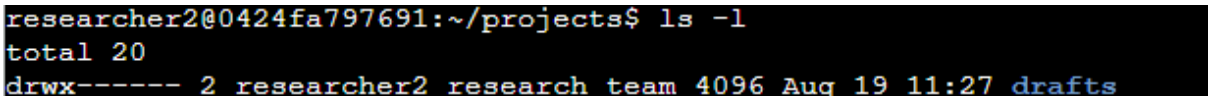
Change directory permissions

Question -> The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents. Use a Linux command to modify the permissions accordingly.

Finally the group should haven't access to the draft directory so we remove the group permissions using **chmod g-x drafts**



And the result should like this.



Summary

This project demonstrates the fundamental importance of Linux file permissions in ensuring system security and protecting sensitive information. Practical exercises involved using key commands such as `ls -l` (or `ls -la` for hidden files) to list and verify current permissions in detail. Additionally, the `chmod` command was used to modify access rights for users, groups, and others. Specific examples were implemented—such as restricting write access to project files and managing visibility and execution permissions for directories like "drafts"—to ensure that only authorized personnel retain the appropriate level of access.